

MOTIVATION:

For those who are constantly losing their belongings, especially on hikes or camping trips, our goal is to create an easily transferrable computer vision software. When something gets lost, it should only take a group of robots a little while to find it.

CURRENT PROBLEMS

- Current computer vision software focuses primarily on object detection, without emphasis placed on secondary features.

OUR SOLUTION

- A computer vision software that can be easily installed and/or adapted on multiple robots.
- The ability to identify certain objects based on qualitative descriptors such as color, shape, and size.

TECH BASIS

- Wireless connectivity (server)
- Obstacle avoidance via lidar
- Object recognition using neural networks

CURRENT LIMITATIONS

Hardware:

- Only tested on the TurtleBot4
- Connectivity is only available over wifi

Software:

- Object recognition is limited to a few objects
- Full utilization of the software is hardware dependent

HOW IT WORKS

1. The user sends a terminal-based command (i.e. "Find a cone.")
2. Robot searches designated area for object, using mapping and pathing software.
3. Using computer vision, the robot detects the object specified in the command
4. The robot sends a ping over wi-fi back to the terminal indicating that an object matching the target was found.



POSSIBLE FUTURE DEVELOPMENT

- A wider variety of object sensitivity
- Development on the hardware to traverse rougher terrain and possibly other avenues of connection other than wi-fi
- Command & control for more than one robot simultaneously